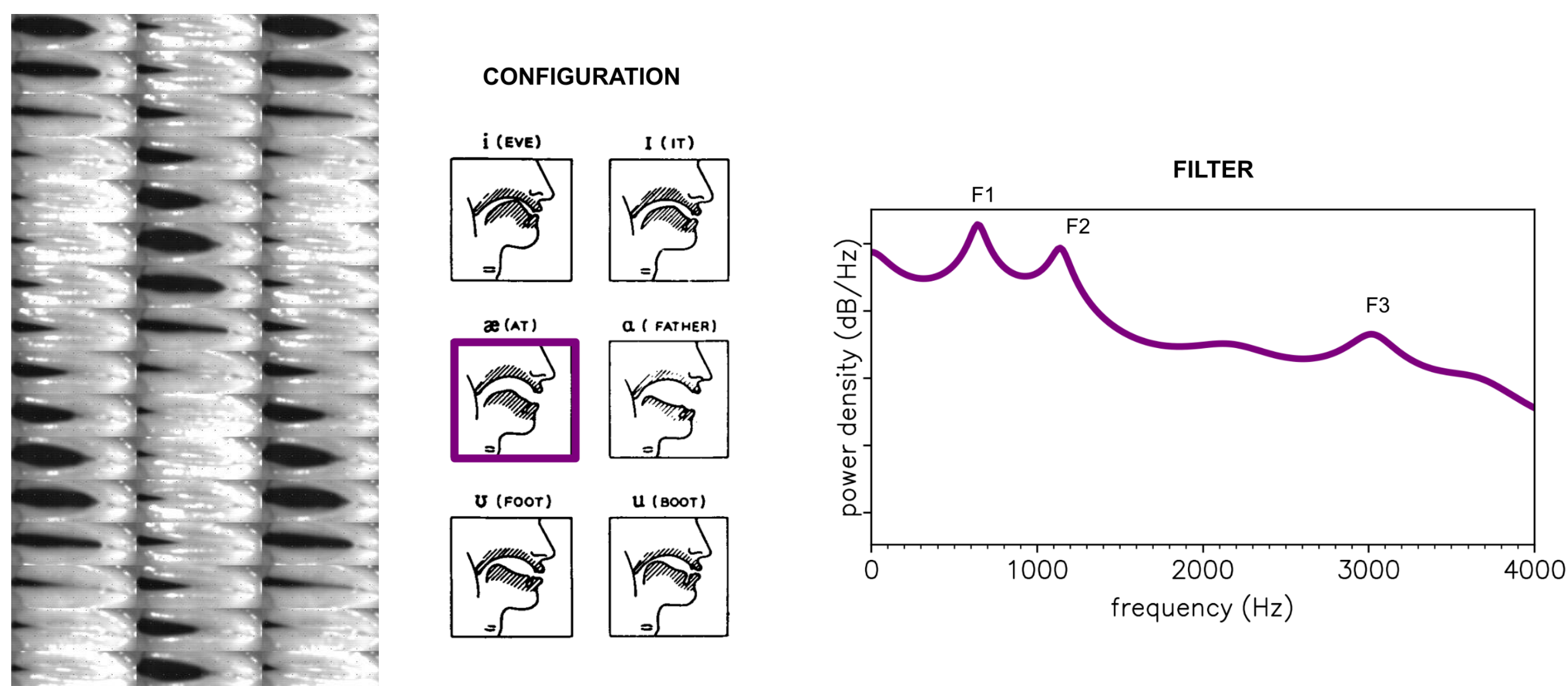


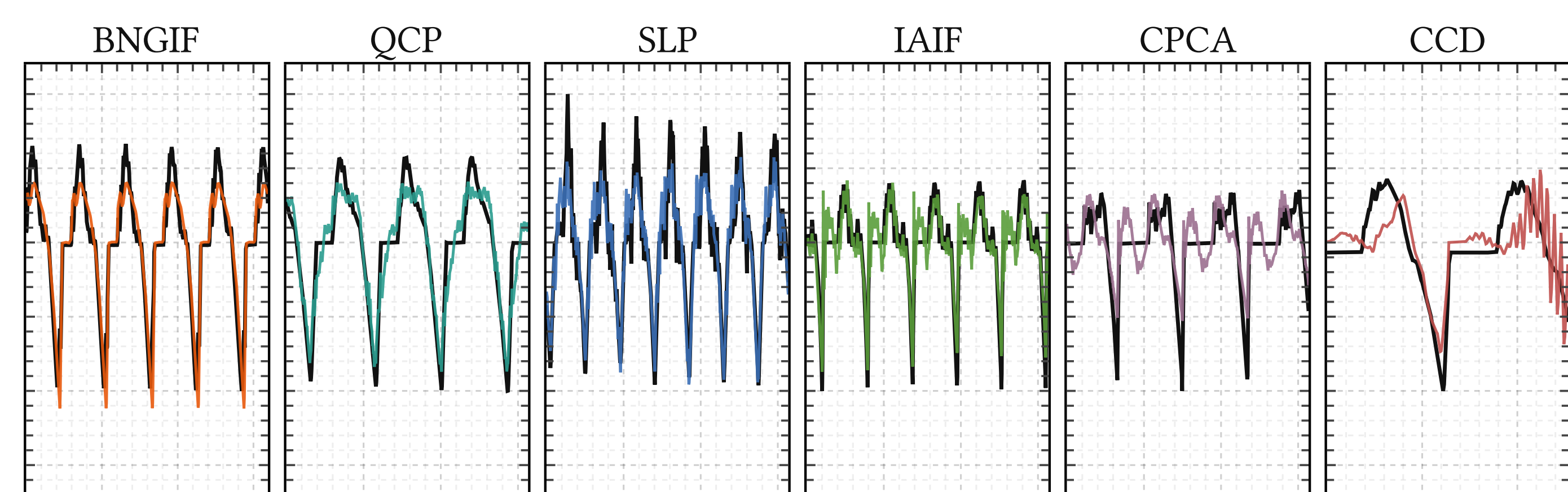
PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

SPEECH = SOURCE × FILTER

The vocal fold airflow and the vocal tract, from the microphone alone.



BEATS THE STATE OF THE ART



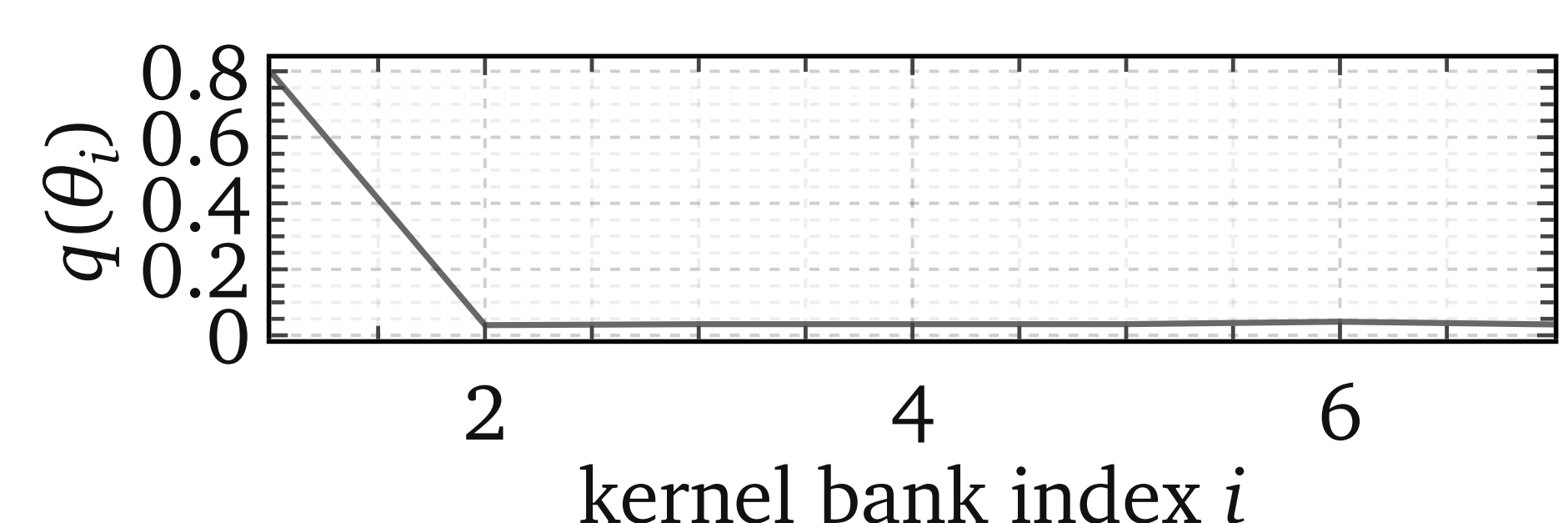
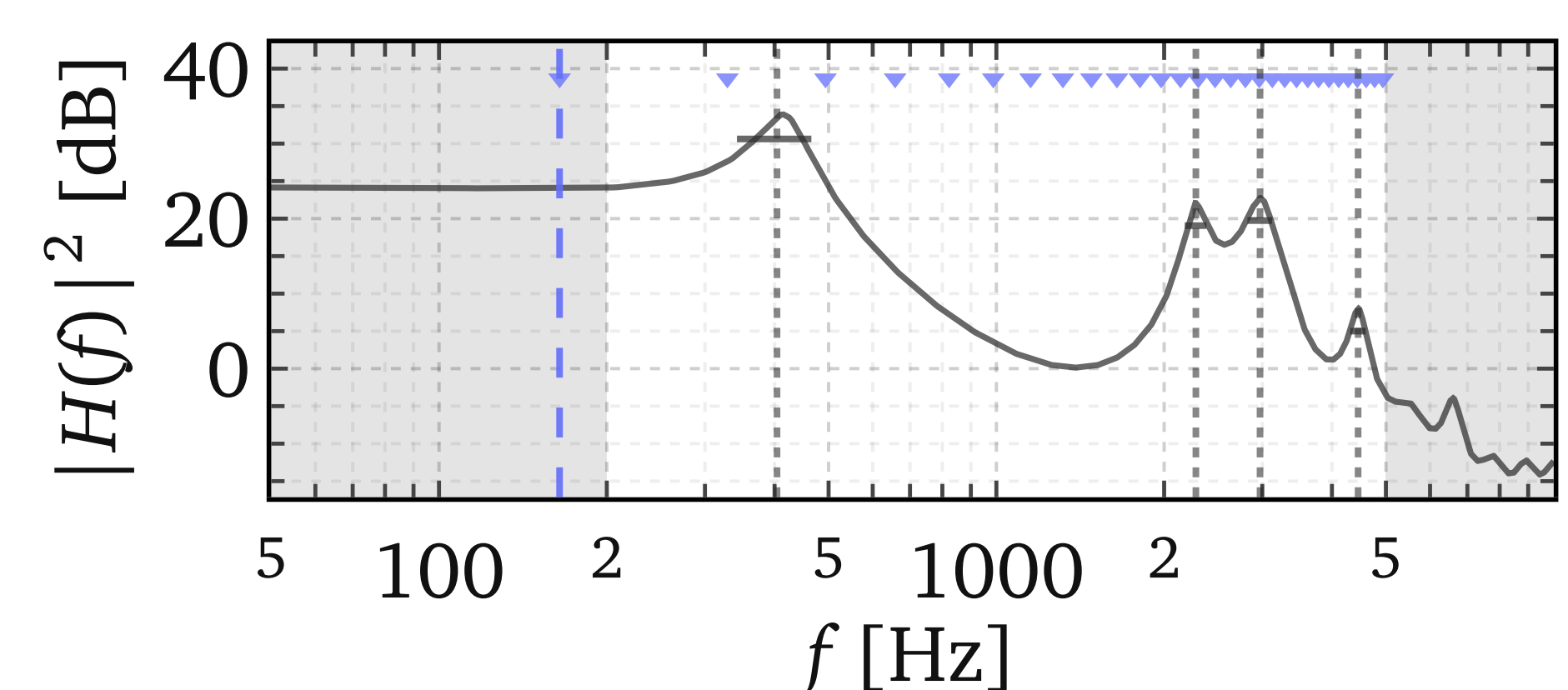
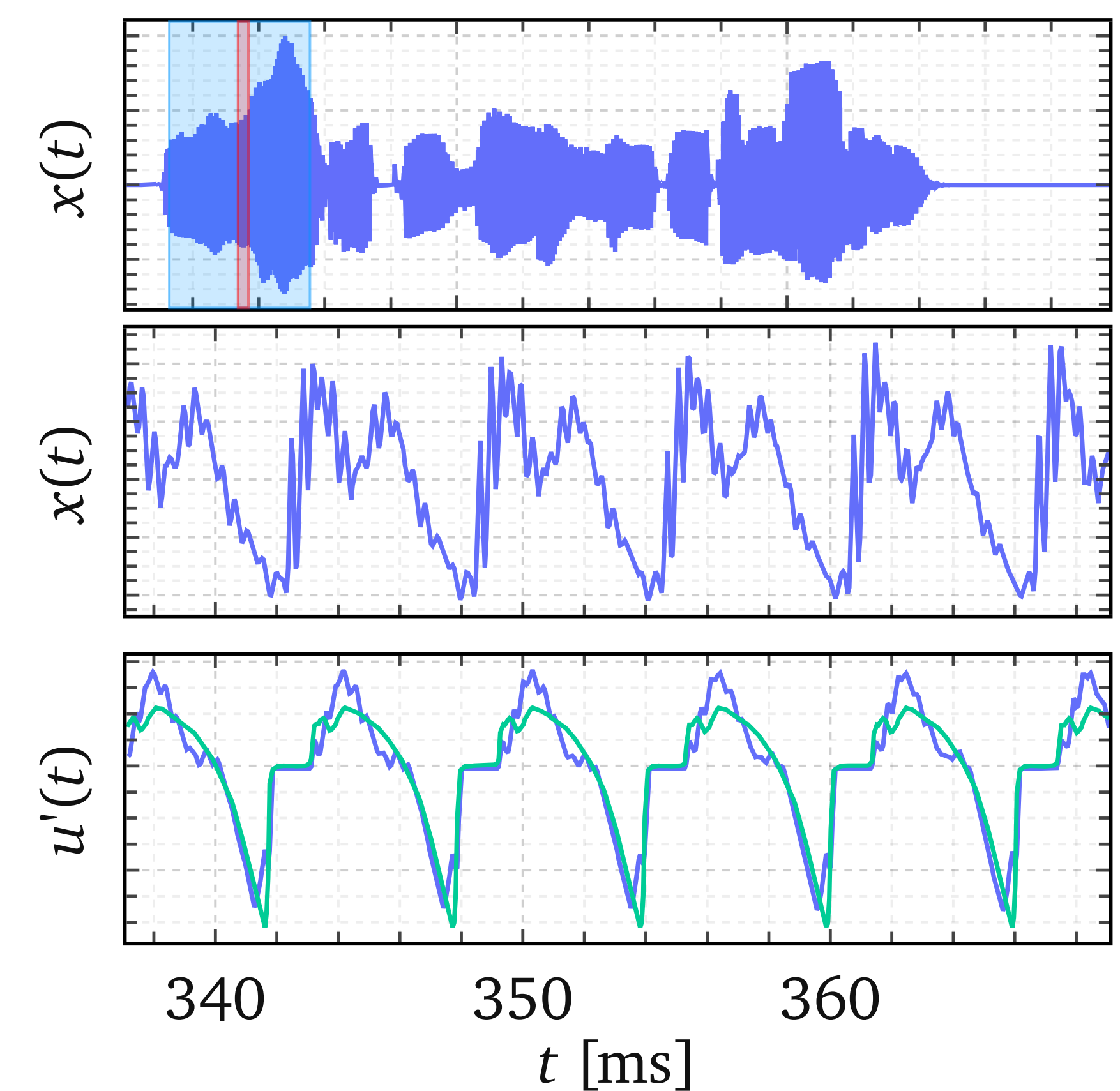
True airflow (black) and six methods' estimates, one frame of the EGIFA benchmark.

	BNGIF	QCP	SLP	IAIF	CPCA	CCD
BNGIF	0.935	<.001	<.001	<.001	<.001	<.001
QCP	1.000	0.910	<.001	<.001	<.001	<.001
SLP	1.000	1.000	0.889	<.001	<.001	<.001
IAIF	1.000	1.000	1.000	0.886	<.001	<.001
CPCA	1.000	1.000	1.000	1.000	0.759	0.004
CCD	1.000	1.000	1.000	1.000	0.996	0.741

Over the whole benchmark: better than each of them, $p < .001$.

ONE MODEL, ALL THE UNCERTAINTY

Posterior over the airflow (band), the filter, and which prior shape explains the frame.



5× FASTER THAN REAL TIME

A second of speech in 0.2 s: variational inference in JAX, one GPU.

WHY IT MATTERS

Hearing implants – could pitch and formants *with error bars* improve sound coding?

Clinics – physiological voice biomarkers, not black-box features.

Phonetics, forensics – formant tracks 30% more accurate than Praat.

